

SECTION 07 31 13

ASPHALT SHINGLES

13FEB2020

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM D1970/D1970M	(2015a) Self-Adhering Polymer Modified Bituminous Sheet Materials Used as Steep Roofing Underlayment for Ice Dam Protection
ASTM D226/D226M	(2009) Standard Specification for Asphalt-Saturated Organic Felt Used in Roofing and Waterproofing
ASTM D3018/D3018M	(2011) Standard Specification for Class A Asphalt Shingles Surfaced With Mineral Granules
ASTM D3161/D3161M	(2016a) Standard Test Method for Wind-Resistance of Steep Slope Roofing Products (Fan-Induced Method)
ASTM D3462/D3462M	(2016) Standard Specification for Asphalt Shingles Made From Glass Felt and Surfaced with Mineral Granules
ASTM D41/D41M	(2011; R 2016) Standard Specification for Asphalt Primer Used in Roofing, Dampproofing, and Waterproofing
ASTM D4586/D4586M	(2007; E 2012; R 2012) Asphalt Roof Cement, Asbestos-Free
ASTM D4869/D4869M	(2016a) Standard Specification for Asphalt-Saturated Organic Felt Underlayment Used in Steep Slope Roofing
ASTM D6380/D6380M	(2003; E 2013; R 2013) Standard Specification for Asphalt Roll Roofing (Organic Felt)
ASTM D7158/D7158M	(2016) Standard Test Method for Wind Resistance of Asphalt Shingles (Uplift Force/Uplift Resistance Method)

NATIONAL ROOFING CONTRACTORS ASSOCIATION (NRCA)

NRCA 0418

(2009) Steep-slope Roof System Manual

U.S. DEPARTMENT OF ENERGY (DOE)

Energy Star

(1992; R 2006) Energy Star Energy
Efficiency Labeling System (FEMP)

UNDERWRITERS LABORATORIES (UL)

UL 2218

(2010; Reprint May 2012) UL Standard for
Safety Impact Resistance of Prepared Roof
Covering Materials

UL 790

(2004; Reprint Jul 2014) Standard Test
Methods for Fire Tests of Roof Coverings

1.2 DEFINITIONS

1.2.1 Top Lap

That portion of shingle overlapping shingle in course below.

1.2.2 Head Lap

The triple coverage portion of top lap which is the shortest distance from the butt edge of an overlapping shingle to the upper edge of a shingle in the second course below.

1.2.3 Exposure

That portion of a shingle exposed to the weather after installation.

1.3 SUBMITTALS

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Shingles

Submit data including type, weight, class, UL labels, and special types of underlayment and eave flashing.

Information Placard

SD-04 Samples

Shingles

Full shingle sample and manufacturer's standard size samples of materials and products requiring color or finish selection.

Color Charts: Must be submitted to Camp Lejeune BEAP representative (Base Architect) for approval.

SD-08 Manufacturer's Instructions

Application

SD-11 Closeout Submittals

Manufacturer's Warranty

Contractor's Warranty

1.4 DELIVERY AND STORAGE

Deliver materials in the manufacturer's unopened bundles and containers bearing the manufacturer's brand name. Keep materials dry, completely covered, and protected from the weather. Store according to manufacturer's written instructions. Store roll goods on end in an upright position or in accordance with manufacturer's recommendations. Immediately before laying, store roofing felt for 24 hours in an area maintained at a temperature not lower than 10 degrees C 50 degrees F.

1.5 WARRANTIES

Warranties must begin on the date of Government acceptance of the work.

1.5.1 Manufacturer's Warranty

Furnish the asphalt shingle manufacturer's standard 25 year warranty for the asphalt shingles. The warranty must run directly to the Government.

1.5.2 Contractor's Warranty

Provide warranty for 5 years that the asphalt shingle roofing system, as installed, is free from defects in workmanship. When repairs due to defective workmanship are required during the Contractor's warranty period, the Contractor must make such repairs within 72 hours of notification. When repairs are not performed within the specified time, emergency repairs performed by others will not void the warranty.

PART 2 PRODUCTS

2.1 MATERIALS

2.1.1 Shingles

Mineral granule-surfaced asphalt shingles, self-sealing, square tab, strip, fungus-resistant. ASTM D3018/D3018M, Type I, and ASTM D3462/D3462M, architectural shingles weighing not less than 11.3 kilograms per square meter 250 pounds per 100 square feet. Shingles must meet the fire resistance requirements of UL 790 for Class A and the wind resistance requirements of ASTM D7158/D7158M, Class H. Color must be as selected from the manufacturer's standard color charts. Shingle color must be submitted to in accordance with Camp Lejeune BEAP representative (Base Architect) for approval.

2.1.2 Roof Sheathing

Plywood: 1/2" thick over entire roof, C-D Grade, Exposure 1, with an Identification Index of not less than 24/0.
Provide exterior grade material with phenol resin for all applications.

2.1.3 Flashing

Wall or Parapet flashing and counterflashing, chimney step flashing, chimney pan, vertical surfaces, etc. : Refer to drawings for materials. Copper (16 oz.) and/or Aluminum (.025")

2.1.4 Self-Adhering Membrane Underlayment

Self-adhering rubberized asphaltic membrane, a minimum of 1 mm 40 mils thick, high temperature as recommended by the manufacturer for use over entire roof and flashing conditions. Products shall meet or exceed the performance criteria of ASTM D1970/D197M.

2.1.5 Nails for Applying Shingles

Aluminum or hot-dipped galvanized steel or equivalent corrosion resistant with sharp points and flat heads 10 to 11 mm 3/8 to 7/16 inch in diameter. Shank diameter of nails must be a minimum of 2.67 mm 0.105 inch and a maximum of 3.43 mm 0.135 inch with garb or otherwise deformed for added pull-out resistance. Nails must be long enough to penetrate completely through or extend a minimum of 20 mm 3/4 inch into roof deck, whichever is less, when driven through materials to be fastened.

2.1.6 Asphalt Roof Cement

ASTM D4586/D4586M, Type II.

2.1.7 Asphalt Primer

ASTM D41/D41M.

2.1.8 Ventilators

2.1.8.1 Nailable Plastic Shingle Over Type Ridge Vents

Ridge vents must be constructed of UV stabilized nailable rigid polypropylene material, approximately 0.30 m 1 foot wide and 25 mm 1 inch thick, and must be in 1.2 m 4 foot long interlocking sections with self-aligning ends or corrugated polyethylene rigid roll or rigid strip ridge vent with aluminum wind deflectors on each side. Vents must be designed to prevent infiltration of insects, rain, and snow.

2.1.8.2 Nailable Mesh Shingle Over Type Ridge Vents

Ridge vents must be constructed of UV stabilized nailable polyester mesh material, approximately 0.30 m 1 foot wide. Vents must be designed to prevent infiltration of insects, rain, and snow. Fasten with 3" nails at each rafter, no exceptions.

2.1.8.3 Information Placard

Information placard must be 8.5 inch x 11 inch and contain the information contained in Table I. Information on the placard must be typed in Courier New 12 pitch. Manufacturer's and Contractor's information must include a phone number. Placard must be able to be mounted on the inside of a building. Placard can be aluminum or acrylic metal or a sheet of paper

inside a standard compatible suitable frame.

PART 3 EXECUTION

3.1 VERIFICATION OF CONDITIONS

Do not install building construction materials that show visual evidence of biological growth.

Ensure that roof deck is smooth, clean, dry, and without loose knots. Roof surfaces must be firm and free from loose boards, large cracks, and projecting ends that might damage the roofing. Vents and other projections through roofs must be properly flashed and secured in position, and projecting nails must be driven flush with the deck.

3.2 SURFACE PREPARATION

Cover knotholes and cracks with sheet metal nailed securely to sheathing. Flash and secure vents and other roof projections, and drive projecting nails firmly home.

3.3 APPLICATION

Apply roofing materials as specified herein unless specified or recommended otherwise by shingle manufacturer's written instructions or by NRCA 0418.

3.3.1 Underlayment

Provide for roof slopes 1 in 3 4 inches per foot and greater. Apply one layer of shingle underlayment to roof deck. Lay underlayment parallel to roof eaves, starting at eaves. Provide minimum 50 mm 2 inch head laps, 100 mm 4 inch end laps, and 150 mm 6 inch laps from both sides over hips and ridges. Nail sufficiently to hold until shingles are applied. Turn up vertical surfaces a minimum of 100 mm 4 inches.

Provide for roof slopes between 1 in 6 2 inches per foot and 1 in 3 4 inches per foot 1 in 34 inches per foot and greater. Apply two layers to roof deck. Provide a 480 mm 19 inch wide strip as starter sheet to maintain specified number of layers throughout roof. Lay parallel to eaves, starting at eaves. Provide minimum 480 mm 19 inch head laps, 150 mm 6 inch laps from both sides over hips and ridges, and 300 mm 12 inch end laps in the field of the roof. Nail sufficiently to hold until shingles are applied. Turn up vertical surfaces a minimum of 100 mm 4 inches. When a self-adhering membrane is used for eave flashing, start underlayment from upper edge of eave flashing.

3.3.2 Drip Edges

Provide metal drip edges as specified in Section 07 60 00 FLASHING AND SHEET METAL applied directly on the wood deck at eaves and over the underlayment at rakes. Extend back from edge of deck a minimum of 75 mm 3 inches, and secure with nails spaced a maximum of 100 millimeters 4 inches o.c. along inner edge.

3.3.3 Starter Strip

Use a row of shingles with tabs removed and trimmed to ensure that joints

are not exposed at shingle cutouts. Apply starter strip along eaves, overlaying and finishing even with lower edge of eave flashing strip overhanging the metal drip edge at eaves and rake edges 6 to 10 mm 1/4 inch to 3/8 inch]; fasten in a line parallel to and 75 to 100 mm 3 to 4 inches above eave edge. Place nails so top of nail is not exposed in cutouts of first course of shingles. Seal tabs of first course of shingles, eaves, edges, and rakes with asphalt roof cement as specified below.

3.3.4 Shingle Courses

Start first course with full shingle, and apply succeeding courses with joints staggered at thirds or halves. Butt-end joints of shingles must not align vertically more often than every fourth course. Apply shingle courses as follows:

- a. Fastening: Do not drive fasteners into or above the factory-applied adhesive unless adhesive is located 16 mm 5/8 inch or closer to top of cutouts. Place fasteners so they are concealed by shingle top lap and penetrate the head lap.
- b. Shingles applied with nails: Nominal 125 mm 5 inch exposure. Apply each shingle with minimum of (6) six nails. Place one nail 25 mm 1 inch from each end, and evenly space nails on a horizontal line a minimum of 16 mm 5/8 inch above top of cutouts. [Cement each tab with one spot of asphalt roof cement placed 25 to 50 mm 1 to 2 inches from bottom edge of shingle.
- d. Sealing: Seal each tab with continuous, 225 mm 9 inch long, 6 mm 1/4 inch diameter bead of asphalt roof cement, applied to the surface of course below. Place bead on horizontal line 16 mm 5/8 inch above cutouts so bead will be 25 mm 1 inch from bottom edge of tab to be sealed and so bead will not show through cutouts. After nailing each shingle, press tabs down to ensure spreading and bonding of asphalt roof cement.

3.3.5 Hips and Ridges

Form with 225 by 300 mm 9 by 12 inch individual shingles or with 300 by 300 mm 12 by 12 inch shingles cut from 300 by 900 mm 12 by 36 inch strip shingles. Bend shingles lengthwise down center with equal exposure on each side of hip or ridge. Lap shingles to provide a maximum 125 mm 5 inch exposure, and nail each side in unexposed area 140 mm 5-1/2 inches from butt and 25 mm 1 inch in from edge.

3.3.6 Valleys

Provide woven valleys.

3.3.6.1 Woven Valleys

Provide valley lining as specified for woven cut valley. Lay valley shingles over lining by either of the following methods:

- a. Method I: Apply regular shingles on both roofs simultaneously. Weave each course in turn over the valley. Lay the first regular course of

shingles along eaves of roof up to and over valley. Extend course along adjoining roof deck at least **300 mm 12 inches**. Carry first regular course of shingles of adjoining roof over valley on top of previously applied shingles. Lay succeeding courses alternately, weaving valley shingles over each other for full length of valley.

- b. Method II: Apply regular shingles on each roof surface separately to a line about **900 mm 3 feet** from center of valley, and weave valley shingles in place later, as specified for Method I.

In following either method, press shingles tightly into valley, and fasten in normal manner; except apply nails not closer than **150 mm 6 inches** to valley centerline, and apply additional nail in top corner of terminal shingle on both sides of valley.

3.3.7 Flashing

3.3.7.1 Eave Flashing

Provide for roof slopes between **1 in 6 and 1 in 3 2 inches per foot and 4 inches per foot 1 in 34 inches per foot** and greater.

- b. From the eaves to **the roof peak** apply one layer of self-adhering membrane **underlayment**. Follow membrane manufacturer's printed installation instructions.

3.3.7.2 Stepped Flashing

For sloping roofs which abut vertical surfaces, provide stepped metal flashing as specified **herewith**.

3.3.7.3 Vent and Stack Flashing

Apply shingles up to point where vent or stack pipe projects through roof, and cut nearest shingle to fit around pipe. Before applying shingles beyond pipe, prepare flange of metal pipe vent flashing as specified **herewith**. Apply a **3 mm 1/8 inch** thick coating of asphalt roof cement on bottom side of flashing flange. Slip flashing collar and flange over pipe, and set coated flange in **2 mm 1/16 inch** coating of asphalt roof cement. After applying flashing flange, continue shingling up roof. Lap lower part of flange over shingles. Overlap flange with side and upper shingles. Fit shingles around pipe, and embed in **2 mm 1/16 inch** thick coating of asphalt roof cement where shingles overlay flange.

3.3.7.4 Chimney Flashing

Provide treated wood crickets. Provide metal base and counterflashing as specified **herewith**. Uniformly coat masonry surfaces which are to receive flashing with asphalt primer applied at rate of **4 liters per 10 square meters 1 gallon per 100 square feet**. Apply shingles over underlayment up to front face of chimney. Apply metal front base flashing with lower section extending at least **100 mm 4 inches** over shingles. Set base flashing in a **2 mm 1/16 inch** coating of asphalt roof cement on shingles and chimney face. Apply metal step flashing at sides in a coating of asphalt roof cement. Embed end shingles in each course that overlaps step flashing

with asphalt roof cement. Apply metal rear base flashing over cricket and back of chimney in coating of asphalt roof cement. Apply end shingles in each course up to cricket, and cement in place. Lap base flashing minimum of 75 mm 3 inches with metal counterflashing.

Information Placard

Coordinate with Engineering Technician location of plaque inside the building.

TABLE I - ROOF INFORMATION PLACARD
DO NOT MAKE ALTERATIONS OR REPAIRS TO THIS ROOF WITHOUT APPROVAL FROM OWNER
Camp LeJeune, NC
Bldg: _____
Contract No: _____ Task Order _____

Manufacturer's Warranty:

Expires: ____ ____ ____
Warranty No: _____
Provided By: Company
Address
Phone

Contractor's Warranty:

Expires: ____ ____ ____
Warranty No: _____
Provided By: Company
Address
Phone

Description of Roof System:

Deck:
Ice and Water Membrane:
Shingle:
Ridge Vent:

-- End of Section --